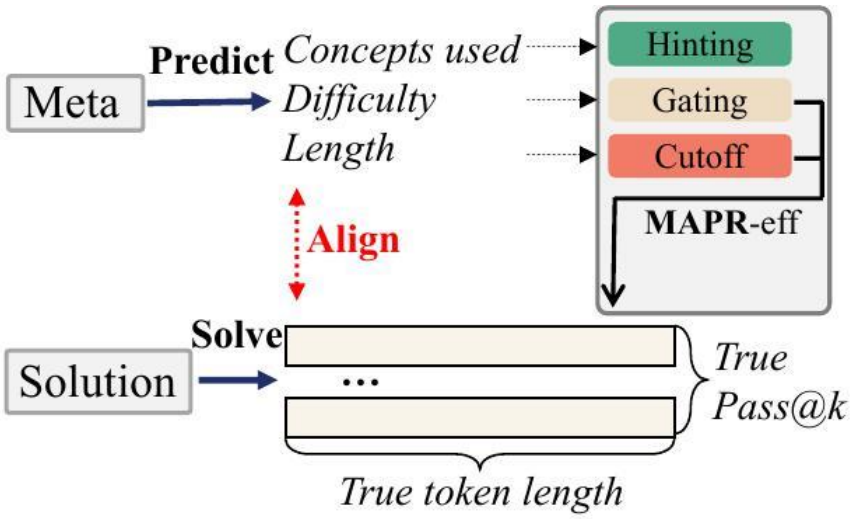
MAPR: Overall Framework

Overview

MAPR predicts and solves in parallel from given meta and solution prompts. Predicted values are verified against true Pass@k, token length, and used concepts from solution rollouts. MAPR-efficient applies predictive gating and length cutoff for efficient training.



① Parallel Predict & Solve

Meta prompt → predicts difficulty, length & concepts used. Solution prompt → generates rollouts simultaneously.

② Alignment via Verification

Predictions are aligned against true Pass@k, true token length, and concepts extracted from solution rollouts.

③ MAPR-efficient Mechanisms

- **Hinting** — concept-guided reasoning hints
- **Meta Gating** — predictive difficulty gating
- **Length Cutoff** — predicted length as rollout budget

Key Insight: RL-trained reasoning models lack self-awareness of how to think. MAPR explicitly fosters meta-awareness — enabling the model to anticipate difficulty, budget token length, and leverage relevant concepts before solving.